# Equations

## Section A

**QUESTION 1**

(a) Solve the following:

(1) $(x-2)(x+1)=0$ (2)

(3) $(x+2)(x-3)=6$ (4)

(b) Solve for $x$ and $y$ simultaneously:

$$2x+3y=10$$
$$x-4y=-6$$ (5)

**[17]**

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## Section B

Solve the following equations:

a) $x-3=\frac{18}{x}$ (4)

b) $\frac{2x-5}{x+3}-\frac{1}{4-x}=\frac{7}{x^2-x-12}$ (6)

c) $-2<2-x\leq 2$, represent your answer on a number line. (3)

## Section C

**QUESTION 1**

Solve for $x$:

a) (1) $2x^2-7x=-6$ (3)

(2) $\frac{x}{x+2}+\frac{x}{3-x}=\frac{3x-2}{x^2-x-6}$ (4)

(3) $5bx=3ax+d$ (2)

b) (1) $-2<2x-2\leq 8$ (3)

(2) Hence, give all the possible solutions to the inequality when it is further given that $x\in\mathbb{Z}$. (2)

**QUESTION 2**

a) Solve for $x$ and $y$ simultaneously if; (5)

$3x + 2y = 29$ and

$x - 3y + 5 = 0$

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## Section D

**QUESTION 1**

(a) Solve for $x$:

(3) $x^2 - 3x - 4 = 0$ (2)

(4) $(x - 2)^2 = 4$ (3)

(b) Solve for $x$ and $y$ simultaneously:

$3x - 4y = 5$

$2y - x = -1$ (5)

**[15]**

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## Section E

**QUESTION 1**

(a) Solve for $x$:

(1) $(x + 2)(x - 3) = 6$ (4)

(3) $t = \frac{y}{5}\sqrt{\frac{p}{x}}$ (3)

(b) Write down the value of the greatest integer that satisfies this inequality

$-6 \leq -2x - 6 < 4$ if $x \in \mathrm{Z}$ (4)